Fuzzy Logic Enabled Advanced Monitoring for Airliner Maintenance

Reka Patel[1[0009−0001−7596−7131]], Hugo Henry[2[0009−0001−2530−3521]], and Dr. Kelly Cohen[3[0000−0002−8655−1465]]

University of Cincinnati, Cincinnati OH 45221, USA
patel9rk@mail.uc.edu henryho@mail.uc.edu
cohenky@ucmail.uc.edu

Abstract. Aircraft advanced monitoring and predictive Maintenance are key objectives of the research aiming to streamline maintenance, improve efficiency, reduce turn around time (TAT) and cut costs. In order to follow the predicted increase of 44% in air traffic by 2050, the need to reduce this down time due to minor maintenance is increasing and we can see a silver lining through applied, trustworthy Artificial intelligence (AI). This paper focuses on minor fault detection in the position and movement of the moving parts of the body of an airliner digital twin. Through this paper we see how fuzzy rule-based systems (FRBS) can be used effectively within the realm of advanced monitoring and predictive maintenance. It also examines how the system could be implemented and it's impact in the maintenance chains such as planning and organization of the maintenance. Other topics such as certification processes for AI systems and training will be briefly touched on. Hence, this paper is a concept presentation of an FRBS for anomaly detection and mild maintenance decision making to streamline non-critical degradations upon actuator systems. Through a dual simulation process comparing a nominal behavior under commands and an actual behavior coupled through a Fuzzy Inference System (FIS), we identify possible faults in order to redirect to the Part M personnel (airworthiness crew) to adapt the maintenance plans and mitigate wear and tear. This concept can be extended to any system equipped with sensors for monitoring and an established maintenance plan.

Keywords: Safety · Avionics · Predictive maintenance · Advanced monitoring · AI · Explainable AI · Responsible AI · Fuzzy Systems.

1 Introduction / Problem Statement

The commercial aviation industry plays an important role in the US economy by transporting people and driving demands for various goods. However, the industry faces many challenges and the amount of air traffic is increasing with flights being added each year. With this increase, certain issues are becoming more prominent in the industry. Some of these issues include aging aircraft and components, technological advancements, and staffing shortages, which increase the

difficulty to maintain both safety standards and operational efficiency. Current practices rely on regularly scheduled inspections, physical checks, and manual troubleshooting. These methods work, but are not as effective for predicting future maintenance and optimizing efficiency.

Aircraft inspections are essential for identifying and fixing any problems that could compromise the aircraft's performance and safety. The main maintenance programs used by airlines are based almost entirely on performance. The FAA defines preventative maintenance as "simple or minor preservation operations and the replacement of small standard parts not involving complex assembly operations" [4]. In other words, preventative maintenance are regularly scheduled checks and service. Current daily basic aircraft maintenance checks include pre-flight, in-flight, and post-flight checks and monitoring. Pre-flight maintenance checks include a walk-around of the exterior and a manual cockpit inspection conducted by the pilots. In-flight monitoring is done by monitoring sensors in real time to gather data. General post-flight checks overlap with pre-flight checks by including physical checks of the exterior and cockpit with additional cleaning. These regular checks aim to prevent minor issues from escalating. In addition to these basic checks in between flights, there are more in depth inspections that occur daily, every few days, and weekly.

These current maintenance checks are all performed physically and take time for the pilots to go through each step. These checks are also focused on simply checking if things work and are not focused on prediction. All of these regular checks are reaction based and don't plan towards keeping the aircraft in flight condition longer. With a growing rate of air traffic there is a need to keep aircraft in good flight condition longer. Additionally, with an increase in staffing shortages there is a need to create a more efficient and streamlined maintenance process. With fast computational times and verifiable results, AI can aid in reducing turn around times for aircraft. The addition of an AI system would streamline the process by reducing the time spent on manual checks. More specifically, fuzzy systems can be used to find discrepancies and classify them linguistically into degradation modes. This would make the outputs more interpretable for personnel and easier to understand. Fuzzy systems also provide traceable decisions allowing them to be validated. Another main concern is maintaining safety. These aircraft carry people and goods thereby making safety an important factor. Having a system that is providing verifiable decision making will maintain the priority of safety.

2 Methodology

This paper is a presentation of a concept mixing a literature survey and an on terrain survey. The literature survey is extracted from the documentation of the authorities (FAA, EASA) and the on terrain survey was completed through feedback on the concept by industry active personnel for realism, proofreading and additions.

3 Background

3.1 Current Aircraft Maintenance Practices

Different types of regular aircraft inspections include pre-flight, daily, periodic, and special. Pre-flight inspections are conducted before every flight to ensure the aircraft is in airworthy condition. Pilots, or a maintenance team, conduct these inspections by walking around the aircraft for a visible inspection of the exterior and checking the functionality of the interior equipment and systems. Some of these components include the airframe, control surfaces, landing gear, cockpit instruments, communication systems, fuel quantity, and navigation systems. This process focuses on seeing if there are any visible issues. Daily inspections, or "turnaround inspections", are typically conducted at the beginning of every day. They cover critical areas such as the landing gear, brakes, engine oil levels, and hydraulic systems. They consist of visual and manual checks of these systems looking for signs of damage or leaks. Periodic inspections are more comprehensive and are conducted at regular intervals ranging from every few days to weekly or biweekly. They contain a detailed examination of the airframe, engines, landing gear, control systems, electrical systems, and avionics. In addition to the physical inspection, the team will also go over previous maintenance logs to help see the patterns of the aircraft's faults. Special inspections are reactive in response to incidents, modifications, or changes to the operating conditions. Some examples of when these would occur are after lighting storms, following a big repair, or after the aircraft has been grounded for an extended period of time. They are more pointed and will usually focus on a specific area rather than the whole aircraft [2]. Additionally, there are different classes of maintenance checks outside of those listed above. Class A checks are the most basic and are performed regularly. They typically occur every 400-600 hours or every 200-300 flights and last about a day. These checks include detailed inspections on the exterior of the aircraft, fluid levels, brakes, avionics systems, and flight controls. Class A inspections would contain most of the aforementioned regular inspections. Class B used to be a more elaborate version of class A occurring every six to eight months however, most airlines have phased these out. Class C checks are more in depth maintenance checks occurring every 18 to 24 months. The aircraft is taken out for one to four weeks and extensive checks are performed on all critical components. Class D checks are the longest and most intensive. They occur every six to ten years and the aircraft gets dismantled for inspection. It takes several weeks, worn pieces are replaced, and the goal is to bring the plane back to its optimal condition [1]. These checks are a huge cost for companies and nowadays they try to get away from this system as much as they can through the use of better tools and more streamlined maintenance. With the advance in technological developments there are now also in-flight sensor monitoring systems. These systems will monitor sensor data in real time to show if there are any deviations from the norm. They provide useful data to show where the aircraft may be having issues however, they currently lack the ability to classify it into a specific degradation mode. While these systems are structured and ef-

fective they do not help predict future issues or help know the actual health of the aircraft. They rely heavily on identifying visible faults and detecting existing problems, not emerging faults. This requires a lot of physical time and trust in human detection. They do not analyze the real-time data or have enough focus on utilizing the trends of the data to predict issues. Additionally, strict scheduled maintenance plans are not efficient in giving the aircraft maintenance when it is needed. Two aircraft may have had two very different days of flying yet they both receive identical maintenance. This is not efficient and why airlines should shift to predictive maintenance rather than reactive based. It is also worthwhile to note that even though we talked about preventive maintenance, most companies rely more on corrective maintenance if not forced by the Maintenance Planning Document (MPD) created by the manufacturer. Lastly, one caveat to corrective maintenance is that large amounts of stock pieces for repairs need to be on hand. In the current industry, most airline companies rely on the interchangeability of the equipments in order to have enough stock for the full fleet and often cannibalize equipments from older aircraft. This interchangeability of equipments is validated by the manufacturer and authorities, but do not need to be told to the Maintenance, Repairs and Operation (MRO) team in charge of the checks. This causes loss of time and human error due to the differences in maintenance between two interchangeable parts.

3.2 Why Fuzzy Systems for Maintenance

In aircraft maintenance operations, decisions flow through a cascading architecture of personnel and teams that receive and interpret maintenance orders across multiple levels. Here, fuzzy logic becomes valuable because not only is it a reasoning tool, but a way to translate complex system data into linguistically meaningful terms that people can understand. The use of linguistic variables enables a transparent communication between the system and human operators. Because the system uses a Fuzzy Rule Based System (FRBS), the decisions, or assessments, made by the system can be recreated and with traceable rule sets. Therefore, every maintenance decision can be followed and audited. This allows full runbacks of prior states and fault progression. Rather than creating a black-box model, the fuzzy approach mirrors the actual flow of maintenance responsibilities thereby enhancing interpretability [13]. Through this, fuzzy systems act as a connection between real-time diagnostics and operational decision making by supporting validation and corrective action.

When it comes to aircraft sensor data it can be noisy, incomplete, or uncertain, so degradation is often hard to discern. Rather than detecting a single failure mode, fuzzy logic can find these hard to discern discrepancies and express them in linguistic terms. It's ability to transform uncertain data into interpretable and traceable decisions is part of what sets it apart from other methods. Additionally, maintenance teams need safe, trustworthy, verifiable, and explainable tools. If properly designed, each output and decision can be audited and reconstructed to show why certain decisions were made. This fosters liability-level assurance, offering trust in automated diagnostics. This trustworthiness

will allow it to comply with the explainability and interpretability necessary for the safety standards of aircraft maintenance [14]. Such a system can be made offline as well as online given the short inference time (order of magnitude of the millisecond). This characteristic allows for different applications and avoidance of certification processes if needed by taking the system from airborne to offline, or the inverse, to provide reactive decision making. Due to the possibility of an offline implementation, we can also directly mitigate redundancy by having the system duplicated as much as we want for comparisons and also mitigate the subject of bit corruption in electronic systems due to electromagnetic flare and cosmic rays. These both are shown to be very important subjects by the recent airbus incident on flight control systems [15].

4 Concept

We are proposing a concept consisting of an advanced monitoring solution utilizing FRBSs. The system will aim to find minor degradations in the aircraft actuators and not in the Minimal Equipment List (MMEL), thereby rendering it not dangerous for flight conditions. It is also done through the existing sensors and data collections of the aircraft making it non-intrusive to the maintenance or flight personnel crew. In order to avoid the difficulties of certification for airborne systems (DO-178) [11], this concept will be an offline evaluator rather than a runtime system. The FRBS will then use a difference between the data collected and a nominal behavior (established through a digital twin or experiments) in order to classify faults and push that information toward the Part-145 personnel (maintenance crew), the Part-M (airworthiness crew) or CAMO if relevant to the situation [8][9][10]. Figure 1 shows the classification methodology.

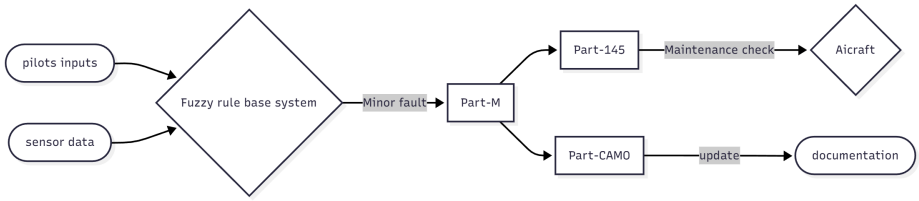

Fig. 1. Concept Diagram

The system will use the data generated during the control surface tests in the pre-flight checklist allowing the nominal behaviors to be known and precise with very few added uncertainties beside degradations, pilot input variations and almost negligible wind effects. For the system design phase, only information accessible via the sensors will be used to make a decision and this will avoid taking unrealistic behavior into the fuzzy system. Some potential inputs to consider for the system include:

- Settling time
- Overshoot
- Sensor bias
- Sensor noise

Settling time will show if the system is modified by external factors such as additional friction or a leak. Overshoot will show coupled degradation. As it is a first order system, correctly controlled, it should not overshoot. If overshoot occurs then a second order mode or a static error would appear yielding a sensor error or delay. Sensor bias would mean an offset in the sensor and therefore, a problem in the health or calibration of it. Finally, the sensor noise is computed as the signal/noise ratio and also reflects the health of the sensor. Failures in sensor performance can lead to delayed control responses, imbalances, or complete failure modes.

From these inputs it yields a multitude of coupled failure modes depending on the configuration taken. One example is to consider the actuators to be hydraulic and the only faults would be increased friction, delay and static error. Therefore, this scenario would yield five degradation modes:

- Abnormal friction
- Possible fluid leak
- Degraded sensor
- Degraded connection
- Sensor actuator generalized degradation

This is just one example of how the logic can be applied. The logic can further be expanded to include more inputs and more failure modes to eventually cover any and all potential issues that could be occurring for the aircraft as stated by the manufacturer. Degradation typically develops gradually, where warning signs could be captured with proper monitoring and classification. Minor discrepancies can also be difficult to detect by visual and manual checks. The proposed solution would provide a verifiable way to convert behavior data into actionable maintenance decisions. It provides an explainable, data-driven monitoring and classification system that fits seamlessly into existing maintenance workflows. This system introduces an offline fuzzy inference system that takes flight data and classifies failure modes. In being able to capture this degradation and take corrective action to lengthen the lifespan of the component, it will aid in lengthening the life of the entire aircraft. Additionally, it will help reduce overall maintenance labor, time and cost by providing more pointed service to the aircraft.

5 Integration and Impact

This concept would be added to the checks that occur in between flights. The maintenance crew (Part-145) would download the monitoring data from the previous flight and input it into the FIS. As there are already systems in place to

monitor in-flight data, this solution would easily integrate into the current maintenance workflow. Based on the data, the FIS would output that there are no issues, a possible failure mode or combined failure modes. Based on this failure mode the maintenance planning team (Part-M) would be able to take a pseudo-preventive action. This action could be a simple fix or cause the team to realize that something is on its way to failure. Over time, patterns will begin to be seen of which parts are degrading more often or sooner than when they are scheduled for maintenance. Based on this data, current maintenance schedules could be amended for optimization. The parts that are degrading faster could have maintenance schedules moved up and, contrarily, parts that are lasting longer could have their schedule pushed back (if allowed by the MPD). Providing corrective maintenance on parts before a full failure occurs will cause less wear and tear on the overall aircraft as other parts are not affected by the one fault. Overall, less wear and tear will allow the aircraft to be in flight condition longer. Additionally, knowing when to fix parts will also increase the aircraft lifespan as there will be less frequent replacements. When looking at making changes to maintenance plans the maintenance planning document (MPD) and aircraft maintenance program (AMP) need to be considered. The MPD, created by the aircraft manufacturer and validated by authorities (FAA, EASA,...), is the origin of the AMP. It provides minimum time between checks and how to perform it through other documentations, such as links with the AMM (Aircraft Maintenance Manual), as well as detailed guidelines for routine maintenance. Airworthiness directives (AD) are mandates issued by aviation authorities to address identified safety concerns. They can modify or introduce new maintenance requirements. Service bulletins (SB) offer recommendations for maintenance practices [7]. They also enable one to say if the service life of a product should be extended by some maintenance or repair [6]. Through the use of service bulletins, recommendations to maintenance practices can be made based on the patterns found in the data with the goal of amending AMPs by the part-M. The AMP receive regular changes by part-M or CAMO based on experience with the aircraft and the company usage. This minor fault monitoring could bring new insights and deeper understanding of why the failures occur. Given that AMP refactors are validated by the authorities, the system needs the auditability of the FRBS to be able to give solid background to the part-M to push an update. It is worth noting that changes in the AMP are not a common occurrence as it concerns mostly heavy maintenance and is costly and lengthy to certify. However, a change in the AMP can have a huge impact in the efficiency and is the core of the maintenance organization. Indeed, in the industry, there are current progressive and preventative maintenance efforts being made through the developments in data collection, analysis techniques and the introduction of AI [12]. For the FAA to monitor safety with new techniques, all systems must comply with their continuing analysis and surveillance system (CASS)[4]. This is a tool to ensure that new programs are safe and efficient. The proposed solution increases safety because it is detecting degradation. Instead of waiting for a full failure, the system is monitoring the precursor of the failure without

having to physically test the system. This allows it to streamline the process of verification to reduce the time between flights thereby reducing the ground time of the aircraft and limiting maintenance cost due to a full replacement of the actuator (costly in pieces and time). This system aims to be a small system akin to agentic systems, very specialized but can be built on top of each other in a cascading way. Therefore, this concept can be extended to any system with sensors and combined with logistics software. The hurdle of training the crew to use the system will be mitigated by the fact that only part-M and CAMO may enter in direct contact with the software. Therefore, the amount of people and resulting cost would be lower than a generalized training for a full maintenance management software such as $AERO - WEBB^{TM}$ [16] that requires heavy investment into formation of the crew, given that the whole organization needs to be trained. Our system would be only a piece that a select few would see. Two other points need to be addressed. Firstly, how our system would be updated. A fuzzy system has the advantage of not needing a full refactor and training if we add or remove a case or membership. We can go and make changes as well as train specific components of the FIS as opposed to a neural network that needs a full training at every change. This allows certifications for the updates to be more easily obtained. Secondly, who would make the update. A specialized individual or team would need to be stationed for continuous surveillance. Same as for other software systems, this is in case of failure or downtime of the architecture. Finally, in the future we need to address a possible but important extension of the concept. This would be the analysis and identification of interchangeability in the system and would get rid of one of the most time consuming factors of verification.

6 Conclusion

As the commercial aviation industry continues to grow, maintenance strategies must adapt to cover the increased demand while maintaining safety. They will need to try to optimize efficiency while following FAA guidelines. This concept proposes an advanced monitoring and corrective maintenance solution utilizing fuzzy inference systems to classify early-stage degradation. This is done by analyzing discrepancies between nominal and actual behavior based on sensor data from the flights. By using this existing sensor data it would integrate easily into current maintenance workflows providing a shift from reactive maintenance to predictive maintenance. Some benefits of this solution are that it is pseudo-predictive rather than purely preventive or corrective in that it reacts to is a precursor of the actual fault. It is also offline, inexpensive to integrate, and doesn't require a major system redesign. The fuzzy logic structure is easy to comprehend and naturally complies with current procedures thereby reducing turn around time between flights. It improves system awareness by preventing minor degradations from progressing or causing full failures, ultimately extending aircraft lifespan. However, some drawbacks are that it needs to comply with regulations, ensure non-tempering on the systems dynamics as well as mainte-

nance related software regulations, and clearly be able to be audited if needed. Moreover, this concept only gives possible failure modes and does not aim to classify actual failure of the system. It's goal is more to identify mild degradations. Despite these limitations, this concept represents a feasible and scalable solution towards trustworthy AI-enabled predictive maintenance. By turning sensor data into interpretable degradation indicators, it enhances system awareness while encouraging safety practices. As air traffic is projected to increase 44% by 2050, a turn towards predictive maintenance will be essential for streamlining efficiency, sustaining long-term reliability, and overall maintenance improvement.

7 Future Work

To expand on this concept, more inputs and failure modes will be created to cover the entire aircraft and a working fuzzy system will be created. The nominal behavior of the actuator sensor couples can be extracted and input into the fuzzy architecture to train it on classifying failure modes. The system can then be checked against the FAA standards and for deployment into current maintenance systems. Additionally, a pseudo documentation and training format will be developed for better realism and an extension of the concept acting as a cascading decision making system will be envisioned. Finally, a full impact analysis and breakdown of the introduction of this new system to an actual team will be done with the help of industry experts and in service maintenance engineers.

Acknowledgments

The authors extend their sincere gratitude to Célian Lavergne for their proofreading and feedback on the realism of the concept as well as deeper insight on the on terrain factors. The authors also extend their thanks to the AI Bio Lab members of the University of Cincinnati for their feedback and continuous support.

References

1. Aircraft Maintenance Checks beyond the Basics `www.acumen.aero/aircraft-maintenance-checks`
2. Aircraft Inspections (Types & Checklist) an In-Depth Guide `Www.acumen.aero, www.acumen.aero/aircraft-inspections`
3. Cordon Oscar. Genetic Fuzzy Systems: Evolutionary Tuning and Learning of Fuzzy Knowledge Bases. Singapore, World Scientific, 2001.
4. Preventive Maintenance, an Industry Solution and Policy Incentive `www.aii.org/preventive-maintenance-an-industry-solution-and-policy-incentive/`. Accessed 24 Nov. 2025.
5. NASA's Gateways to Blue Skies Competition `blueskies.nianet.org/competition/`

6. The Basics of Maintenance in General Aviation https://services.boeing.com/news/aviation-maintenance-basics
7. https://ialta.aero/boeing-an-aircraft-maintenance-program-amp
8. https://www.easa.europa.eu/en/the-agency/faqs/part-145#category-part-145-general
9. https://www.easa.europa.eu/en/the-agency/faqs/part-m
10. https://www.easa.europa.eu/en/domains/aircraft-products/continuing-airworthiness-organisations/part-camo-part-cao
11. https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_20-115D.pdf
12. SAE International : Process Standard for Development and Certification/Approval of Aeronautical Safety-Related Products Implementing AI, ARP6983 (WIP), (2023).
13. The lenses of interpretability: A framework for investigating the interpretability of learned models, Ph.D. dissertation, University of Cincinnati
14. Henry, H., Dogga. B., Ernest, N., and Cohen, K., "Fuzzy logic enabled assured autonomy for air traffic control", Manuscript #56, 2025 Joint IFSA/NAFIPS Conference, August 16-19, 2025, Banff, Alberta, Canada.
15. Airbus AOT A27N022-25 Revision 00 (original issue) https://ad.easa.europa.eu/blob/EASA_AD_2025_0268_E.pdf/EAD_2025-0268-E_1
16. https://2moro.com/aero-webb